Quantitative Macro Theory **HW #5**
Spring 2018 **Due at 5pm on *Friday* 3/2/2018**

Write your answer on latex and prepare a pdf file. The file should be clearly written and organized. You should think of that file as a short version of a paper. The appendix of that file should include your code. Email me that pdf file before the deadline.

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1. (100 points) **Income-Fluctuation problem**

   Consider the problem

$$V(a, y_i) = \max_{c,a'} u(c) + \rho\beta \sum_{j=1}^{3} \Gamma_{i,j} V(a', y_j)$$

   subject to

$$c + a' = Ra + y_i,\ a' \geq \underline{a}.$$

   Calibration:

   - The probability of survival is $\rho = 0.975$.
   - $u(c) = \frac{c^{1-\sigma}}{(1-\sigma)}$, with $\sigma = 2$.
   - The discount rate, $\beta = 0.92$.
   - Gross interest rate, $R = 1.02$.
   - Income is given by y. Use a three-state Markov chain Γ that we loosely identify with "super-rich" ($y_1 = 16,000$), "white-collar" ($y_2 = 80$), and "blue-collar" ($y_3 = 15$).
   - The first state provides the opportunity and incentive for a high concentration of earnings and wealth (see Castañeda, Diaz-Gimenez, and Rios-Rull, 2003). $\Gamma_{2,1} = 0.001$, $\Gamma_{1,1} = 0.02$.
   - The persistence of the latter two states ensures that earnings display a sizable positive autocorrelation: $\Gamma_{2,2} = 0.78$, $\Gamma_{3,3} = 0.964$.
   - This specification requires 6 parameters in the Markov transition matrix but we reduce it to 4 by setting the probability of moving from blue-collar to super-rich and vice-versa to zero. $\Gamma_{3,1} = 0$, $\Gamma_{1,3} = 0$, $\Gamma_{1,2} = 0.98$, $\Gamma_{2,3} = 0.219$, $\Gamma_{3,2} = 0.036$.
   - $\underline{a} = -50$.

   (a) (10 points) Pick your favorite method to solve for $a'(a, y)$ from HW#4 and plot the policy function as a function of a for the three levels of income. Note in the calibration that $\underline{a} = -50$.

   (b) (10 points) Imagine an individual starts with $a = -50$ and "blue-collar" worker income y_3. Think that this individual is 20 years old. Simulate a path of income for this individual for 40 years, until she is 60 years old. Using that path, simulate

the path of assets using $a'(a, y)$. Plot the simulated path of consumption, assets, and income in the same graph. Note that for consumption and assets you should include two paths, one for each value of the borrowing constraint. In the x axis include the age of the individual (each period is a year) starting at 20.

(c) (10 points) Now do the same simulation for 10,000 individuals with the same starting condition $a = 0.5$ and "blue-collar" worker income y_3. Note that the path for income should be different every time. As a consequence, the paths for assets and consumption will be different too. Plot in the y axis the mean of consumption, income, and assets for each age and age in the x axis. Do the same plot for the variance of log consumption and log income.

(d) (15 points) Again with your favorite method, solve for $a'(a, y)$ but setting $\underline{a} = 0$. Do the same simulation you did in (c) and plot in the y axis the mean of consumption, income, and assets for each age and age in the x axis. Do the same plot for the variance of log consumption and log income. If you plot together the cases of $\underline{a} = 0$ $\underline{a} = -50$ your figure at to age 60 should look like (not very close, but with some similar features) this one in Kaplan and Violante (American Economic Journal: Macroeconomics, 2010).

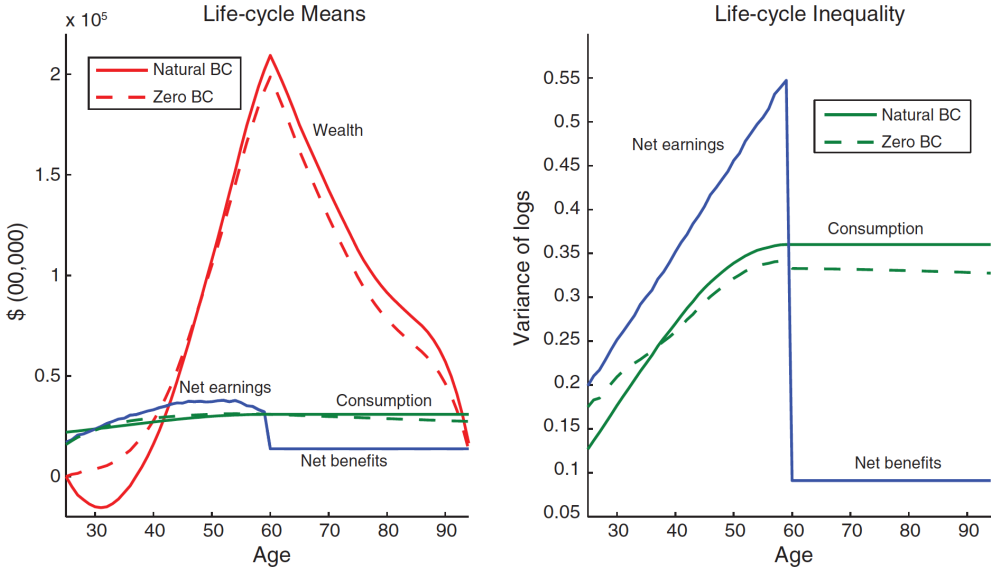


FIGURE 1. LIFE-CYCLE PROFILES FOR MEANS AND VARIANCES IN THE NBC AND ZBC ECONOMIES

(e) (15 points) Kapan and Violante also compute the "insurance coefficients" implied by the model. The insurance coefficient for income shocks in our framework are

$$\phi^{\Delta y} = 1 - \frac{cov(\Delta y_{i,t}, \Delta c_{i,t})}{var(\Delta y_{i,t})} \tag{1}$$

If consumption and income changes co-variate perfectly, there is no insurance. Note that you can compute this covariance and variance with your simulated data and obtain $\phi^{\Delta y}$. What value do you obtain for the cases of $\underline{a} = 0$ and $\underline{a} = -50$? How does it compare with Kaplan and Violante (2010) finding (note that you have only

one shock, and they have permanent and transitory shocks).

TABLE 1—RESULTS FROM THE BENCHMARK MODELS WITH NBC AND ZBC

	Permanent shock			Transitory shock		
	Data BPP	Model BPP	Model TRUE	Data BPP	Model BPP	Model TRUE
Natural BC	0.36 (0.09)	0.22	0.23	0.95 (0.04)	0.94	0.94
Zero BC	0.36 (0.09)	0.07	0.23	0.95 (0.04)	0.82	0.82

(f) (10 points) Note that you can also compute those variance and covariance for a given age. Thus, one can obtain an insurance coefficient for each age. Do it for the cases of $\underline{a} = 0$ $\underline{a} = -50$ for each age from 20 to 60 years old. Plot it with the insurance coefficient in the y axis and age in the x axis. This is how it looks in Kaplan and Violante (2010).

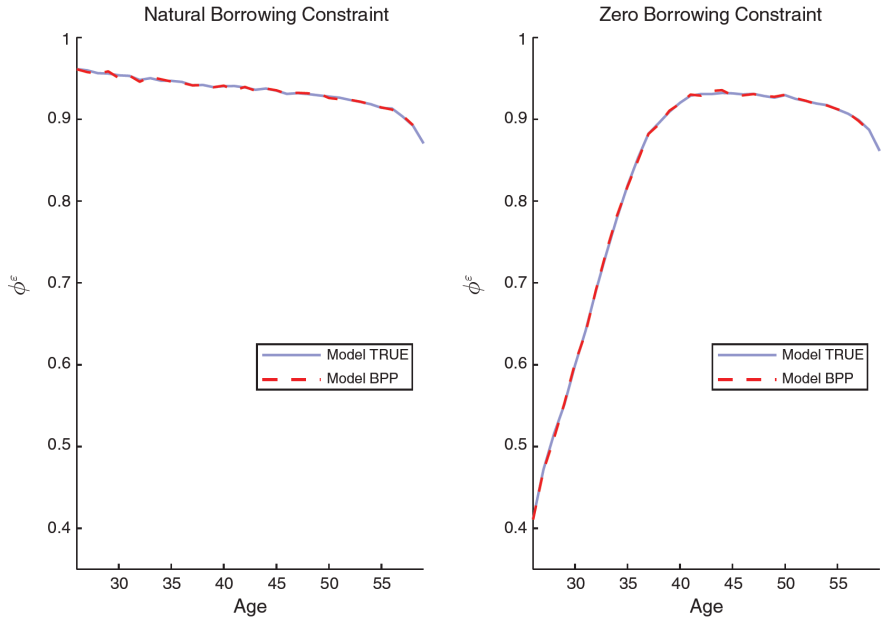


FIGURE 2. AGE PROFILES OF INSURANCE COEFFICIENTS FOR TRANSITORY SHOCKS

(g) (15 points) For the case of $\underline{a} = -50$, keeping $R = 1.02$, and using the method of discretization and interpolation of the density function f discussed in class, solve for the stationary distribution of assets and income. Plot an histogram of income and another one for assets. What is the aggregate stock of assets in the economy?

$$A = \sum_{i \in \{1,2,3\}} \int_{\underline{a}}^{a_{max}} a f_i(a) da \tag{2}$$

where i stands for income level.

Not that among other parameters, this stock of assets depends on R, so we will write $A(R)$. In this case what you found is $A(1.02)$.

(h) (15 points) Solve the policy function $a'(a, y)$ and the density function f for $R = 1.04$. Compute the aggregate stocks of assets for this new level R, $A(1.04)$. Do the same for $R = 1$, $R = 1.06$, and $R = 1.075$. Note that now you have found $A(1), A(1.02), A(1.04), A(1.06), A(1.075)$. Plot this with R in the x axis and $A(R)$ in the y axis. What would you guess is the value of R such that $A(R) = 0$? Why?